| INCH-POUND |

MIL-B-28686
11 September 1990
SUPERSEDING
MIL-B-23997A(YD)
11 August 1981

MILITARY SPECIFICATION

BOILERS, HOT WATER, WATERTUBE, (BENT TUBE, MULTI-DRUM AND CROSS DRUM) PACKAGED TYPE (320,001 TO 125,000,000 BTU/HR THERMAL OUTPUT CAPACITY)

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1. Scope. This specification covers watertube (bent drum, multi-drum and cross drum) packaged hot water generator units.

1.2 Classification. The hot water generator shall be one of the following types, sizes, and classes (see 6.2).

TYPES

Type V - Hot water generator for low temperature water not exceeding 160 pounds per square inch gage pressure (psig) or 250 degrees Fahrenheit (^{o}F).
Type VI - Hot water generator for high temperature water exceeding 160 psig or 250°F.

SIZES

Size - 2 - Thermal output capacity between 320,001 and 10,000,000 British thermal units per hour (Btu/hr).

| Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. |

AMSC N/A FSC 4410

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:38Z
Check the source to verify that this is the current version before use.

Size - 3 - Thermal output capacity between 10,000,001 and 35,000,000 Btu/hr.
Size - 4 - Thermal Output Capacity between 35,000,001 and 125,000,000 Btu/hr.

CLASS

Class 1 - Heavy oil fired
Class 2 - Light oil fired
Class 3 - Gas fired
Class 4 - Combination heavy oil and gas fired
Class 5 - Combination light oil and gas fired
Class 6 - Coal fired
Class 7 - Coal/oil fired
Class 8 - Coal/gas fired
Class 9 - Coal/oil/gas fired

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standard. The following specifications and standard form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-B-3180 - Boilers and Related Equipment, Packaging of.
MIL-B-18796 - Burners, Single: Oil, Gas, and Gas-Oil Combination for Packaged Boilers (320,000 to 125,000,000 Btu/hr Thermal Output Capacity).
MIL-S-28683 - Stokers, Mechanical: For Packaged Boilers Fired on Coal or a Combination of Coal-Gas, Coal-Oil, or Coal-Gas-Oil (2,000,000 to 50,000,000 Btu/hr Thermal Output Capacity).
MIL-B-28685 - Baghouse, Reverse Pulse Jet; for Packaged Boilers Fired on Coal or a Combination of Coal and Gas, Coal and Oil, or Coal-Gas-Oil (2,100,000 to 125,000,000 Btu/hr Thermal Output Capacity).

STANDARD

MILITARY

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:38Z
Check the source to verify that this is the current version before use.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which are current on the date of the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Boiler and Pressure Vessel Code.
Section I - Power Boilers.
Section IV - Heating Boilers.
Section VIII - Pressure Vessels.
ASME CSD-1 - Controls and Safety Devices for Automatically Fired Boilers.
ASME Performance Test Code.
PTC 4.1 - Steam Generating Units

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA ICS 1 - General Standards for Industrial Controls and Systems.
NEMA ICS 2 - Industrial Control Devices, Controllers and Assemblies.
NEMA ICS 6 - Enclosures for Industrial Controls and Systems.
NEMA MG 1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W. Washington, DC 20037.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.
NFPA 85A - Explosion Prevention in Fuel Oil- and Natural Gas-Fired Single Burner Boiler-Furnaces.

(Application for copies should be addressed to the National fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

UNDERWRITERS LABORATORIES, INC. (UL)

UL 353 - Limit Controls.

(Application for copies should be addressed to the Underwriters Laboratories, Inc. 333 Pfingsten Road, Northbrook, IL 60062.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:38Z
Check the source to verify that this is the current version before use.

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The hot water generator shall be a packaged watertube style with a bent tube, multi-drum or cross-drum design. The hot water generator shall be capable of producing the hot water output capacity as specified (see 6.2). The hot water generator shall be fired on the fuel specified (see 6.1.1 and 6.2). The hot water generator shall be furnished complete with the burner specified in MIL-B-18796 or the stoker specified in MIL-S-28683) or both as specified (see 6.2). Expansion or pressure regulating vessels, water circulating pumps, feed water regulating valves and associated controls are not considered part of this document. Each hot water generator shall be furnished complete with all appurtenances specified herein, and shall be factory wired and ready for operation when connected to water, fuel, and electrical supplies.

3.2 Codes and standards. The hot water generator package shall conform to the applicable requirements of Section I, Section IV, and Section VIII of the ASME Boiler and Pressure Vessel Code (hereinafter referred to as the ASME Code), ASME CSD-1, NFPA 70, and NFPA 85A. Limit controls and interlock switches shall be in accordance with UL 353.

3.3 Standard commercial product. The hot water generator shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the hot water generator being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3.1 System of measurement. The dimensions used in this document are not intended to preclude the use of the metric system of measurement in the fabrication and production of the material, individual parts, and the finished product, provided form, fit, and function requirements are satisfied.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification shall be new.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:38Z
Check the source to verify that this is the current version before use.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Design. The hot water generator shall be designed to withstand the normal strains, jars, and vibrations incidental to shipping, storage, and installation. The hot water generator shall be designed for the following operating parameters as specified (see 6.2):

a. Design pressure (psig).
b. Operating pressure at hot water generator outlet (psig).
c. Design temperature (^{o}F).
d. Operating temperature at hot water generator outlet (^{o}F).
e. Maximum continuous thermal output (Btu/hr).
f. Return temperature (^{o}F).
g. Maximum and minimum anticipated ambient air temperatures (^{o}F).
h. Hot water generator site elevation above sea level (feet).

3.6.1 Stoker retrofit design. When specified (see 6.1.1 and 6.2), the design of the hot water generator shall be such as to allow the hot water generator to be retrofitted with a stoker if a stoker is not currently required. Hot water generators designed to be retrofitted shall have maximum flue gas velocities and furnace exit temperatures in the range typical of other coal fired boilers. Tubes in the convection section shall be designed to be in line. Stoker fired hot water generators shall have a thermal output capacity of between 2,000,000 and 50,000,000 Btu/hr. If the maximum thermal capacity of the stoker fired hot water generator is less than the thermal output capacity of 3.6, then the instructions shall also include how supplemental oil or gas burners shall be used to attain the required thermal output capacity of 3.6. Supplemental oil or gas burners shall be limited to 40 percent of the required thermal output capacity.

3.6.2 Thermal shock. The hot water generator shall be designed to withstand the thermal stresses associated with cold start and emergency shutdown operating conditions.

3.7 Performance characteristics.

3.7.1 Thermal efficiency. When tested in accordance with 4.7.2, the thermal efficiency shall be not less than 81 percent for light and heavy oil fuels, 77 percent for gas fuel, and 80 percent for coal fuel. The efficiency requirements are independent of an auxiliary heat recovery unit and are the lowest they can be at any firing rate.

3.7.2 Furnace heat input rate. When the hot water generator is operating at maximum output, the heat input rates to the furnace shall be not greater than the values specified in 3.7.2.1 and 3.7.2.4 as applicable.

3.7.2.1 Gas and oil fired equipment volume heat input rate. The volume heat input rate for gas and oil fired hot water generators shall not be greater than 64,000 Btu/hr per cubic feet (cu ft) of furnace volume.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:38Z
Check the source to verify that this is the current version before use.

3.7.2.2 Coal fired equipment volume heat input rate. Unless otherwise specified (see 6.2), the volume heat input rate for coal fired hot water generators shall not be greater than 30,000 Btu/hr per cu ft except for firing lignite or sub-bituminous coal. Hot water generators firing sub-bituminous coal shall have a volume heat input rate not greater than 45,000 Btu/hr per cu ft (see 6.1.2).

3.7.2.3 Surface heat input rates. The maximum heat input rate in Btu/hr per square foot (sq ft) of effective radiant heating surface of the furnace shall be not greater than the values specified in table I.

TABLE I. Maximum allowable heat input rate.

Rated capacity, thousand Btu/hr	Heat input rate Btu/hr per sq ft
less than 30,000	80,000
31,001 to 61,000	100,000
61,001 to 92,000	115,000
92,001 to 125,000	125,000

3.7.2.4 Stoker fired boiler grate heat release rates. The maximum heat release rates per square foot of effective air admitting grate area shall not be greater than the values specified in table II.

TABLE II. Maximum allowable stoker grate heat input rates.

Stoker Design	Maximum Grate Heat Release Rate Btu/hr/sq ft
Spreader Stoker	700,000
Overfeed Stoker	450,000
Underfeed Stoker	450,000
Fixed grate stoker	360,000

3.7.3 Exit temperature at stack. The exit temperature of the flue gas at the stack shall be not less than 325°F without an economizer. When exit temperatures of not less than 325°F without an economizer are allowed, the flue gas passages shall be of corrosion resistant materials to prevent potential flue gas condensate corrosion problems.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:38Z
Check the source to verify that this is the current version before use.

When operating at minimum load and an economizer is required (see 3.16), the exiting flue gas temperature shall be not less than 300°F. The economizer water temperature at flue gas exit shall not be less than 220°F. When exit flue gas temperatures of less than 300°F are allowed use corrosion resistant materials in the flue gas passages and economizer to prevent potential flue gas condensate corrosion problems.

3.7.4 Baghouse design data requirements. When specified (see 6.2), the boiler manufacturer shall provide the following information to be used for the design and installation of a flue gas baghouse in accordance with MIL-B-28685.

a. Maximum gas glow in standard cubic feet per minute.
b. Required draft at boiler flue gas outlet in inches of water at maximum gas flow conditions.
c. Dimensions of flue gas outlet.
d. Height of flue gas outlet above grade and relative orientation.
e. Flue gas outlet particulate loading. –
f. Flue outlet temperature.

3.8 Environmental requirements. The emission requirements shall be met at the maximum required continuous output. The hot water generator shall meet the legal emission requirements required by the local, state, and Federal environmental rules and regulations. The emission requirements for opacity, particulates, nitrogen oxides, carbon monoxide, and sulfur dioxide shall be as specified (see 6.1.1 and 6.2).

3.9 Construction. The hot water generator shall be constructed for the design conditions and performance requirements specified herein and in accordance with the applicable sections of the ASME Code requirements herein.

3.9.1 Drums. If furnished, drums shall be of adequate size to allow maintenance access to any internals. Inspection ports shall be provided in the casing to permit inspection of lower drum handhole plates while the hot water generator is firing. No parts of a drum or header surface shall be directly exposed to furnace radiation unless it is adequately protected by a refractory or a water screen in such a manner that any part of the exposed drum or header surface shall be more than 2 inches from the nearest water tube.

3.9.2 Tubes. Tubes shall be either electric-resistance welded or seamless steel. Tube bends shall be performed in a manner to permit cleaning of the internal surface of tubes. Unless otherwise specified (see 6.2), radiant tube clearances shall be not greater than 1 inch with membrane or finned tube construction. Spinners, turbulators, and other such devices shall not be permitted in the tubes.

3.9.3 Flow distribution. The method of flow distribution may include the installation of restriction orifices on each parallel tube pass, limiting the number of parallel passes, or using forced jet internal circulation or natural circulation with adequate tube size and design.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:38Z
Check the source to verify that this is the current version before use.

3.9.4 Water pressure drop. The water pressure drop shall be designed for the hot water generators to operate at maximum required capacity. Flow restrictions causing temporary pressure drops shall not cause cavitation or flashing of the hot water when the generator is operated at the required temperature and pressure.

3.9.5 Refractories. Refractories shall withstand the temperature existing under maximum load conditions. The refractories shall be formed in place or cast in sections of such size as to be easily replaceable through existing openings. The refractory sections shall be designed and secured in position in such a manner as to withstand the vibration and shocks occurring during shipment.

3.9.6 Access doors and observation ports. The hot water generator shall be equipped with a refractory-lined access door for internal inspection and repairing of the furnace fireside. The hot water generator shall be equipped with cover plates or handholes to allow for internal inspection and cleaning. Not less than two heat resistant observation ports shall be furnished to allow observation of combustion conditions and flame pattern. Observation ports shall be fitted either with safety shields to maintain closure unless held open by the operator, or with air-cooled heat resistant glass closures. Access to the furnaces may be through the burner opening provided it has an unrestricted opening of not less than 15 inches in diameter. Access doors and observation ports shall be gas-tight.

3.9.7 Baffles. Baffles, as required, shall be arranged to diffuse the gas properly over heating surfaces in order to obtain maximum heat absorption. The baffles shall be held securely in place without being affected by or interfering with the free expansion or contraction of the hot water generator. If the baffles are steel they shall be not less than 0.25 inch thick or of heat resistant alloy not less than 0.125 inch thick. Tangent tube baffles may be used in lieu of metal plate baffles. Baffles shall withstand the temperature under maximum load conditions. The partition wall between the furnace and boiler bank shall be of membrane wall construction if the remainder of the furnace is of similar construction.

3.9.8 Furnace. The walls and floor of the furnace shall permit free expansion of the hot water generator without undue stress on any part of the hot water generator. The furnace side walls and roof shall be of water-cooled type construction for units not fired on coal. Water wall headers and supply tubes shall be protected from the products of combustion. Unless otherwise specified (see 6.2), the furnace shall be not greater than 35 feet in length.

3.9.9 Insulation. All exterior surfaces of the hot water generator, including the floor, but excluding doors, drumheads, hand holes, manholes, vents, and appurtenances, shall be insulated and jacketed. The insulation shall be so formed and secured in place as to prevent sagging or displacement during shipment and operation. The insulation shall be covered with a sheet metal jacket made of steel having a thickness of not less than 0.023 inch (24 Standard Gage) or of nonferrous sheet metal having a thickness of not less than 0.029 inch (22 Standard Gage). All sections of the hot water generator exposed to the products of combustion shall be backed by refractory, by insulation, or both, to maintain a casing temperature of not greater than

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:38Z
Check the source to verify that this is the current version before use.

130°F with a surface air velocity of 5 feet per second and an ambient temperature of 80°F while the hot water generator is operating at maximum capacity. No asbestos insulation materials shall be used.

3.9.10 Inner casing. If an inner casing is included in the manufacturer's standard design, it shall be pressure-tight to prevent the products of combustion from coming in contact with the hot water generator insulation and outer casing. The inner casing may either be the tube wall itself or a separate inner casing having a thickness of not less than 0.135 inch (10 Standard Gage), placed just behind the tubes or the refractory backing. The casing shall be tested for tightness at either 1.5 times the design pressure of the weakest section to be tested or 20 inches of water, whichever is less. The pressure drop shall be not greater than 5 inches in ten minutes when tested in accordance with 4.4.

3.9.11 Base. The hot water generator shall be mounted on steel supports fastened securely to a structural steel base. The base shall be capable of supporting the weight of the hot water generator and its contained water.

3.10 Hot water generator trim. Each hot water generator shall be fully equipped with the trim required under the applicable section of the ASME Code for the type of hot water generator specified herein. The additional appurtenances specified herein shall also be furnished. When disassembly is required for shipping clearance and prevention of damage during shipment, such trim and appurtenances may be field installed. The hot water generator trim, valves, connecting piping and fittings shall be suitable for the design pressure and temperature as specified herein. Limit controls and interlock switches shall be in accordance with the applicable requirements of ASME CSD-1, NFPA 85A, and UL 353. Signals shall be compatible with the control system required in MIL-B-18796 or MIL-S-28683. Unless otherwise specified, (see 6.2), the trim and appurtenances shall include but not necessarily be limited to the following:

a. Low water level alarm or flow switch.
b. A top drum vent valve with threaded connection and corrosion resistant trim.
c. A chemical feed pipe connection with a diameter of not less than 0.75 inch with a corrosion resistant trimmed stop valve installed to inject chemical into an actively flowing system return line.
d. A temperature gage on the fluid outlet. This temperature gage may be one of the four gages to be furnished with an economizer when an economizer is to be furnished (see 3.16).
e. An indicating flow meter with dual switches. One to hold the burner out until flow is established the other to take the unit down if flow is lost while running.
f. Outlet Temperature Indicator: operating temperature control - auto reset, excess outlet water temperature safety cutout switch - manual reset, inlet and outlet water pressure gage flash indicator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:38Z
Check the source to verify that this is the current version before use.

3.11 Burners and stokers. Class 1, class 2, class 3, class 4, class 5, class 7, class 8, and class 9 hot water generators shall be furnished with factory installed burners, ignition systems, and associated fuel systems in accordance with MIL-B-18796. Class 6, class 7, class 8, and class 9 hot water generators shall be provided with a stoker in accordance with MIL-S-28683.

3.12 Combination fuel hot water generators. Class 7, class 8, or class 9 hot water generators shall be designed in accordance with all requirements of MIL-S-28683 for the mechanical stoker and all requirements of MIL-B-18796 for the gas, oil, or gas and oil burner. In cases of conflict or duplication of equipment specified, MIL-B-18796 shall take precedence. The systems shall be integrated to operate simultaneously. A common control panel for both the burner and the stoker shall be provided.

3.13 Draft fans and primary air supply systems. The draft fan for hot water generators shall conform to MIL-B-18796. Primary air supply systems for stokers shall conform to MIL-S-28683.

3.14 Combustion control system. The hot water generator shall be furnished with combustion control systems conforming to MIL-B-18796 and MIL-S-28683 as applicable. The firing rate shall be controlled as a function of the hot water generator outlet temperature.

3.14.1 Burner combustion control system. The burner combustion control system shall be in accordance with MIL-B-18796. When specified (see 6.1.1 and 6.2), the combustion control system shall be furnished with oxygen compensation system or an oxygen compensation system with an unburned combustible gas analyzer. The combustion control system shall be one of the following as specified (see 6.2):

a. Fixed rate, on-off control.
b. Three position high-low-off control.
c. Single positioning control.
d. Parallel positioning control.
e. Metering control.

3.14.2 Stoker combustion control system. The stoker combustion control system and control linkage shall be in accordance with MIL-B-28683. The hot water generator manufacturer shall furnish automatic combustion controllers for the stoker control linkages. The controllers shall maintain the fuel and air input in accordance with thermal load demand. The relative rates of change shall be controlled simultaneously so as to maintain the air to fuel ratio within the limits required for continuous combustion and stable furnace pressure throughout the controllable operating range of the stoker.

3.14.3 Combustion control system for combination fuel hot water generators. The combustion control systems for class 7, class 8, and class 9 hot water generators shall be in accordance with 3.14.1 and 3.14.2. The stoker combustion system shall maintain the fuel and air input at or below the manually set limit. The gas and or oil burner thermal input shall be modulated as necessary to supply the additional thermal input to meet the required thermal load demand.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:38Z
Check the source to verify that this is the current version before use.

3.15 Control panel. The controls including operating switches, indicating lights, gages, alarms, motor starters, fuses, and control systems circuits shall be mounted on a single control panel or cabinet as practicable in order to centralize the control functions. The location of the panel shall either be at the side of the hot water generator, or in a free-standing control cabinet for remote mounting as specified (see 6.2).

3.16 Economizer. When specified (see 6.1.1 and 6.2), a separately packaged economizer shall be furnished. The economizer shall be complete with prefabricated interconnecting breeching for attachment to the hot water generator flue gas outlet and to the feed piping between the economizer and the hot water generator inlet and shall require minimum field assembly. The economizer shall be equipped to heat feedwater. An automatic feedwater preheat system shall be provided when necessary to prevent potential corrosion or flue gas condensation. Manual valving shall also be provided to allow the feedwater to bypass around the economizer. The valving and piping shall be designed so that there is no shutoff valve between the economizer outlet and the feedwater inlet to elimiante the need for a separate economizer relief valve. Economizers specified with class 1, class 4, class 6, class 7, class 8, and class 9 hot water generators shall be furnished with soot blowers specified in 3.17. Provisions shall be made for future installation of soot blowers on economizers specified with class 2, class 3, or class 5 hot water generators. The economizer shall be designed and constructed in accordance with the applicable section of the ASME Code. Materials used in construction shall withstand the temperatures and pressures prevailing under maximum load conditions.

3.17 Soot blowers. Soot blower(s) shall be furnished with class 1, class 4, class 6, class 7, class 8, or class 9 hot water generators. Provisions shall be made for future installation of soot blowers on class 2, class 3, and class 5 hot water generators. Soot blowers shall be air operated, and shall be furnished complete with wall sleeves, clamps, hangers, supports, air motor driven and other appurtenances required for a complete installation. The blower elements shall be so arranged that all parts of the heating surfaces shall be cleaned of any soot deposits when rotated by manual or automatic means. Elements shall be of such length, diameter, and total nozzle area that for the operating pressure involved there will be no significant difference in the cleaning effect between the nozzle nearest the inlet and those farthest from the inlet of the element. Elements and material shall be selected for the flue gas temperature anticipated. Soot blower piping shall include a valve chain operated from the floor and drain connection. Soot blowers shall be fitted with scavenging air ports and shall be connected by tubing to a draft source in a manner to provide sufficient air for continuous scavenging of the blowers while the hot water generator is firing during periods when soot blowers are not operated for cleaning purposes.

3.18 Fuel oil preheaters. When specified (see 6.1.1 and 6.2), electric fuel oil preheating systems shall be furnished as part of the packaged hot water generator. The preheater and associated controls shall be located on the hot water generator assembly in addition to the trim heater described in MIL-B-18796. The preheating equipment shall be designed in accordance with Section VIII of the ASME Code. The system shall be designed to heat 120 percent of the maximum fuel oil flow from the specified oil supply temperature

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:38Z
Check the source to verify that this is the current version before use.

(see 6.2) to the required atomization temperature. Pressure drop across the preheater shall be not greater than 15 psi. When the required fuel oil temperature is reached and maintained by the heater, a thermostatic control or limit switch shall cut off the heater. A high temperature preheater cutoff switch shall also be provided with a manual reset. Temperature gages shall be provided to measure fuel oil inlet and outlet temperatures across the preheater.

3.18.1 Electric preheater. When required (see 3.18), the electric preheater shall be furnished with a temperature controller to maintain the fuel oil temperature within 5 percent of the set point temperature. Unless otherwise specified (see 6.2), preheater operating voltage shall be not greater than 230 volts. Electric preheater rated up to 1,800 watts (W) shall be controlled by a line voltage thermostat switch or by a magnetic contactor with a coil operated by a thermostat. Preheaters rated at greater than 1,800 W shall be controlled by a magnetic contactor having a magnetic coil operated by a thermostat. The electric power and separate control circuits shall be protected by a circuit breaker located on the hot water generator assembly.

3.19 Cinder return system. When specified (see 6.2), class 6, class 7, class 8, or class 9 hot water generator shall be furnished with a cinder return system. The cinder return system shall include all necessary conveyor lines, nozzles, ducts, dampers and other equipment for returning the cinders from settling chamber hoppers, under the hot water generator drums, and other internal locations where cinder buildup will occur. The cinder return system shall only use air from the overfire air system. All cinder conveyor lines, elbows, and cinder pick-up nozzles shall have a nominal inside diameter of not less than 3 inches. Materials of construction shall be of abrasion resistant iron.

3.20 Transition piece. Unless otherwise specified (see 6.2), the manufacturer's standard transition piece for adapting the hot water generator flue gas outlet to a vertical free standing stack shall be furnished.

3.21 Electrical requirements. Electrical wiring design practices shall be in accordance with the applicable requirements of NFPA 70. The hot water generator and associated equipment shall obtain power from a circuit with a single manual service disconnect switch and automatic circuit breaker for all hot water generator electrical equipment. Motors shall be in accordance with the provisions of NEMA MG 1, shall be totally enclosed fan cooled and designed for operation on the voltage, phase, and frequency as specified (see 6.2). Unless otherwise specified (see 6.2), motors of 100 horsepower and less shall be provided with across-the-line starter and overload protection. Motors over 100 horsepower shall have reduced voltage starting. Motor starters shall conform to the applicable requirements of NEMA ICS 1 and ICS 2. Motor starters and controls shall be enclosed in NEMA ICS 6 Type 12 enclosures and may be installed in the control panel of 3.15.

3.22 Cleaning, treatment, and painting. Bare steel surfaces shall be cleaned and a primer coat applied. Paint shall be applied to normally painted surfaces in good commercial practice.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:38Z
Check the source to verify that this is the current version before use.

3.23 Lifting and tiedown attachments. The hot water generator shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to type II or type III of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the hot water generator. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown markings shall clearly indicate that the attachments are intended for the tie down of the hot water generator on the carrier when shipped.

3.24 Mounted operating instructions. Unless otherwise specified (see 6.2), operating instructions shall be summarized on a decal or corrosion resistant plate to be mounted inside the control panel door or similar weatherproofed location.

3.25 Identification marking. Identification shall be permanently and legibly marked directly on the hot water generator or on a corrosion-resistant metal plate. Identification shall include manufacturer's name, hot water generator model and serial number, thermal capacity, maximum allowable working pressure, ASME Code stamp, year manufactured, and other pertinent information as identified in the ASME Code. The DoD contract number that this equipment is procured under shall also be marked on the equipment or identification plate.

3.26 Spare parts. When specified (see 6.2), spare parts shall be furnished and shipped with each hot water generator. When furnished, the spare parts required and the quantity thereof shall be as specified (see 6.2).

3.27 Factory start-up service. Unless otherwise specified (see 6.1.1 and 6.2), factory service personnel shall be provided to supervise installation and start-up of the packaged hot water generator for five 8-hour working days.

3.28 Workmanship. Workmanship shall be of such quality as to produce hot water generators in accordance with the ASME Code, as applicable, and shall meet the requirements of this specification and standards prevailing among manufacturers who normally produce this type of equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:38Z
Check the source to verify that this is the current version before use.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.2.1).
b. On-site inspection (see 4.2.2).

4.2.1 Quality conformance. This inspection shall be performed at the manufacturer's facilities. Unless otherwise specified (see 6.2), the quality conformance inspection shall include the examination of 4.3, the inner casing pressure tests of 4.4, the fuel system pressure test of 4.5, the relief valve pressure test of 4.6, and the packaging inspection of 4.8.

4.2.2 On-site inspection. The on-site inspection shall be in addition to the quality conformance inspection performed at the manufacturer's facilities. Unless otherwise specified (see 6.2), the on-site inspection shall consist of the tests of 4.7. All or part of the on-site inspection may be performed at the manufacturer's facilities if specified by the contracting officer (see 6.2). When specified (see 6.2), the examination of 4.3 and either the tests of 4.4 or 4.5 or both, shall be repeated on-site.

4.3 Examination. Each hot water generator and related equipment supplied under this specification shall be examined for compliance with the requirements specified in section 3 of this document. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability.

4.4 Inner casing pressure test. Furnaces and flue gas passages of hot water generators normally operated under pressure shall be tested for tightness at either 1.5 times the design pressure of the weakest section to be tested or 20 inches of water, whichever is less. The section of the unit to be tested shall be carefully sealed using suitable means for blanking any area not to be subjected to test pressure. Air shall be admitted to the unit until the test pressure is reached and then the unit shall be carefully sealed. The outer casing shall be vented to the atmosphere during the casing pressure test. If the pressure drop does not exceed 5 inches of water in ten minutes,

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:38Z
Check the source to verify that this is the current version before use.

the unit shall be regarded as satisfactorily tight and accepted. Failure to meet the requirements of 3.9.10 shall constitute cause for rejection. Any change in air temperature which affects the test conditions shall be taken into consideration.

4.5 Fuel systems pressure test. The preassembled fuel system that is furnished integrally with the hot water generator shall be checked for leakage during the fire testing of the hot water generator. The soap bubble or the halogen-sniff test method shall be used to verify integrity of the gas fuel system.

4.6 Relief valve set pressure test. Relief valves required by the ASME Code shall be individually tested on a test bench using a gas to confirm the required set pressure. The test shall be conducted at a facility that normally performs this work.

4.7 On-site tests. Operational fire tests and auxiliary equipment tests shall be performed on each installed hot water generator. These tests shall be in addition to any manufacturer's standard fire test procedure, unless compliance evidence provided by the hot water generator manufacturer is accepted at the discretion of the contracting officer or his authorized representative. Acceptable fire test results from identical or similar hot water generator equipment installed at the same site may be used at the discretion of the contracting officer as proof that equipment procured with this specification is in compliance with one or more of the tests of 4.7.1 through 4.7.9.

4.7.1 Operational tests. The hot water generator shall be tested to demonstrate control and operational conformance to the requirements of this specification and MIL-B-18796 or MIL-S-28683, as applicable, under varying load conditions ranging from the specified maximum capacity to the minimum turndown ratio without on-off cycling. Unless otherwise specified (see 6.2), oil burners shall be tested for ability to make a cold start from the fuel oil supply temperature specified in 3.18. Automatic burners shall also be tested for ability to provide hot water in accordance with demand when on-off cycling is required. The economizer, when furnished, shall satisfy the temperature requirements of 3.16. Hot water generators that are set up and operated in accordance with manufacturer's recommendations and exhibit (1) excessive or unexplained loss of ignition, (2) nuisance shutdown due to faulty burner or control operation, (3) improper flame, (4) excessive carbon deposits, or (5) necessity for frequent or difficult adjustments shall be rejected.

4.7.2 Capacity and efficiency tests. The capacity and efficiency at the specified maximum capacity of the hot water generator shall be determined in accordance with the abbreviated efficiency test of the ASME PTC 4.1 using the heat loss method. Performance shall be as specified in 3.7. Methods of measuring the flow of water and fuel other than weighing will be allowed. The accuracy of the method shall assure the Government the desired performance specified in 3.7 and be acceptable to the contracting officer or his authorized representatives. Failure to meet the requirements of 3.7 shall constitute cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:38Z
Check the source to verify that this is the current version before use.

4.7.3 Equipment tests. Control panels, preheaters, insulation, valves, controllers and regulators, soot blowers, electric motors, dampers, and other hot water generator related equipment shall be operated as part of the capacity and efficiency tests. The operation of this equipment shall be closely observed during the operational tests for possible defects or nonconformance. The action of dampers, actuators, and valves shall be smooth without backlash.

4.7.4 Sequencing. The hot water generator shall start, operate, and stop in strict accordance with the specified operating sequence.

4.7.5 Burner tests. Tests shall be conducted on the burner as specified in MIL-B-18796 and as specified herein.

4.7.6 Stoker tests. Tests shall be conducted on the stoker as specified in MIL-S-28683 and as specified herein.

4.7.7 Limit and cutout switches. Safety shutdowns shall be caused by simulating fuel safety interlock actuating conditions for each hot water generator limit and cutout switch specified in 3.10. The safety shutdowns shall occur without damage to the equipment.

4.7.8 Drain valves and try cocks. All drain valves and try cocks shall be tested for proper operation.

4.7.9 Emissions testing. The emissions testing for opacity, particulates, nitrogen oxides, carbon monoxide, and sulfur dioxide specified in 3.8 shall be conducted in accordance with the procedures of the governing Federal, state, or local environmental agency as specified (see 6.2).

4.8 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of section 4 of MIL-B-3180.

5. PACKAGING

5.1 Preservation, packing, and marking. The hot water generators shall be preserved, packed, and marked in accordance with MIL-B-3180. The level of preservation and level of packing shall be as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The hot water generators covered by this specification are intended for generation of hot water to be used in processing operations and space heating systems in land installations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:38Z
Check the source to verify that this is the current version before use.

6.1.1 Selection factors. The following should be considered in the selection of a hot water generator and the exercise of options herein:

a. The hot water generator type (oil, gas, coal, or combination) should be determined in accordance with the availability or economics of the fuel supply and any DoD requirements for multiple fuel firing capabilities (see 3.1).
b. Normally, oil-fired hot water generators above 5,000,000 Btu/hr output should be capable of burning fuel oil grades No. 2, No. 4, No. 5, and No. 6. When changing grade of fuel, replacement may be necessary (1) of the fuel nozzles because of the different fuel viscosity, and (2) of the fuel pump motor to drive the pump at a speed that ensures required fuel pressure in order to maintain the required burner output and meet the optimum air/fuel ratio (see 3.1).
c. The expansion vessel, water circulating pump, feedwater regulator and associated controls are not part of this document because this equipment is beyond the scope of a packaged hot water generator skid (see 3.1).
d. Retrofitting a gas or oil fired hot water generator with a stoker will require the front wall to be removed and replaced with a stoker front. A cinder return system may be added if economical. Refractory may be added in back of the bottom portion of the radiant tubes. This provision will add a significant cost to the packaged hot water generator. Vendor should supply equipment to meet air quality regulations. Local environmental regulations and coal availability should be considered before specifying this design requirement (see 3.6.1).
e. If the hot water generator is subject to environmental regulations, each agency should consult with their environmental support office since regulation limits of emissions vary between city, state, and Federal agencies. Vendor should supply equipment to meet air quality regulations (see 3.8).
f. Oxygen compensation system should not be specified for fixed rate on-off, or three position high-low-off combustion controls (see 3.14.1).
g. Economizers can increase fuel efficiencies from 1.0 to 3.0 percent (see 3.16).
h. The minimum water outlet temperature should be 220°F to protect the tubes from corrosion caused by condensing flue gases. A separate mixing system should be installed if a lower hot water temperature is required (see 3.16).
i. A packaged hot water generator mounted preheater should be specified only at installations of single heavy oil fired hot water generators. It is more economic for multiple hot water generator installations to use a common preheater that is external from the individual packaged hot water generator skids (see 3.18).
j. Electric preheaters should be furnished on units firing No. 4 or No. 5 (light) fuel oil if required for proper atomization in low pressure, air atomizing systems (see 3.18).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:38Z
Check the source to verify that this is the current version before use.

k. Hot water generators should either use electric preheaters mounted on the packaged hot water generator or a remote preheater furnished with safety devices to prevent fuel oil form contaminating the circulated hot water (see 3.18).
l. Factory service should be specified only if the hot water generator will be installed and tested without a long period of interim storage (see 3.27).
m. Fungus resistant varnish conforming to MIL-V-173 should be used to coat electrical components and circuit elements, including terminal and circuit connections, when the hot water generator is to be installed in humid conditions. Components and elements inherently inert to fungi or in hermetically sealed enslosures or current-carrying contact surfaces should not be coated.
n. Electromagnetic interference suppression (EMI), when required, should conform to the EMI suppression requirements and test limits for Class C3, Group I equipments as specified in MIL-STD-461. The hot water generator should be subjected to tests to determine conformance in accordance with MIL-STD-462.

6.1.2 Coal characteristics. Lignite or sub-bituminous coal is characterized by lower Btu content by weight and agglomerating tendencies. Nonagglomerating coal with a Btu content less than 11,500 Btu per pound fall in this category.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, date, and number of this specification.
b. Types, sizes, and classes of hot water generator required (see 1.2).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. Required hot water output capacity (see 3.1).
e. Fuel required (see 3.1).
f Whether the hot water generator is to be furnished with a burner as specified in MIL-B-18796 or a stoker as specified in MIL-S-28683 or both (see 3.1).
g. Design parameters for hot water generators (see 3.6):

(1) Design pressure (psig).
(2) Operating pressure at hot water generator outlet (psig).
(3) Design temperature (^{o}F).
(4) Operating temperature hot water generator outlet.
(5) Maximum continuous thermal output (Btu/hr).
(6) Return temperature (^{o}F).
(7) Maximum and minimum anticipated ambient air temperature (^{o}F).
(8) Hot water generator site elevation above sea level (feet).

h. When the hot water generator is not to be designed to allow a retrofit with a stoker (see 3.6.1).
i. Volume heat input rates for coal fired hot water generators, if other than as specified (see 3.7.2.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:38Z
Check the source to verify that this is the current version before use.

j. When the manufacturer is to provide the following information for design and installation of a flue gas baghouse (see 3.7.4):

(1) Maximum gas flow in standard cubic feet per minute.
(2) Required draft a hot water generator flue gas outlet in inches of water at maximum gas flow conditions.
(3) Dimensions of flue gas outlet.
(4) Height of flue gas outlet above grade and relative orientation.
(5) Flue gas outlet particulate loading.
(6) Flue outlet temperature.

k. Emission requirements for opacity, particulates, and nitrogen oxides, carbon monoxide, sulfur dioxide (see 3.8).
l. Width of fins on radiant tubes, if other than as specified (see 3.9.2).
m. Length of furnace, if other than as specified (see 3.9.8).
n. Trim to be supplied with hot water generator, if other than as specified (see 3.10).
o. When the combustion control system should be furnished with oxygen compensation system or oxygen compensation system with unburned combustible gas analyzer (see 3.14.1).
p. Type of burner combustion control system required (see 3.14.1).
q. Location of control panel (see 3.15).
r. When an economizer is required (see 3.16).
s. When an electric fuel preheating system is requested (see 3.18).
t. Fuel oil supply temperature (see 3.18).
u. Operating voltage for electric preheaters, if other than as specified (see 3.18.1).
v. When a cinder return system is required for Class 6, Class 7, Class 8, or Class 9 hot water generators (see 3.19).
w. When the standard transition piece is not to be furnished (see 3.20).
x. Operating power supply for motors (see 3.21).
y. When motors of 100 horsepower and less are not to be provided with across-the-line starter and overload protection (see 3.21).
z. When mounted operating instructions are other than as specified (see 3.24).
aa. When spare parts are required. If required, the type and quantity to provided (see 3.26).
bb. When factory start up service is other than as specified (see 3.27).
cc. Quality conformance inspection, if other than as specified (see 4.2.1).
dd. On-site inspection, if other than as specified (see 4.2.2).
ee. When all or part of on-site inspection may be performed at the manufacturer's facility (see 4.2.2).
ff. Whether the examination of 4.3 and either the tests of 4.4 or 4.5, or both are to be repeated after hot water generator is delivered on-site (see 4.2.2).
gg. When oil burners are not to be tested for ability to make a cold start (see 4.7.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:38Z
Check the source to verify that this is the current version before use.

hh. Agency required procedures for emission source testing (see 4.7.9).
ii. Level of preservation and level of packing required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations (FAR) Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below should be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Reference Paragraph	DID Number	DID Title
6.3.1.1, 6.3.1.2, 6.3.1.3, 6.3.1.4, 6.3.3, 6.3.5 and 6.3.7	MISC-80678	Certification Data/Report
6.3.2 and 6.3.4	DI-E-30115	Installation - Engineering Data
6.3.6	UDI-E-21344	Instruction, Equipment Installation
6.3.8	DI-M-2050	Commercial Manuals

(DIDs related to this specification, and identified in section 6 will be approved and listed as such in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3.1 Standards compliance. Prior to approval of shipment, the contractor should submit to the contracting officer or authorized representative satisfactory evidence that the proposed hot water generator to be furnished under this document meets the requirements of the ASME Code, ASME CSD-1, NFPA 85A, and UL 353.

6.3.1.1 ASME Code requirements. Acceptable evidence of meeting the requirements of Section I, Section IV, and when an economizer or preheater is required, Section VIII, Division 1 of the ASME Code is the ASME code symbol.

6.3.1.2 ASME CSD-1 requirements. Acceptable evidence of meeting the requirements of ASME CSD-1 should be the receipt of a manufacturer's certificate of compliance, stating the equipment procured with this document is in accordance with the applicable requirements of ASME CSD-1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:38Z
Check the source to verify that this is the current version before use.

6.3.1.3 NFPA requirements. Acceptable evidence of meeting the requirements of NFPA 70 and NFPA 85A is the receipt of the manufacturer's certificate of compliance stating the equipment procured with this document is in accordance with the applicable requirements of NFPA 70 and NFPA 85A.

6.3.1.4 UL requirements. Acceptable evidence of meeting the requirements of UL 353 is the certification symbol, or listing mark on the products, or a certified test report from a recognized independent testing laboratory indicating the boiler controls have been tested and conforms to UL requirements.

6.3.2 Stoker retrofitting instructions. The manufacturer should submit instructions for retrofitting the hot water generator with the recommended stoker for converting to coal firing (see 3.6.1).

6.3.3 Design verification. The manufacturer should furnish data to verify compliance with the specified furnace heat input rates (see 3.7.2).

6.3.4 Baghouse installation. Installation instructions covering the connection of the flue gas baghouse should be provided (see 3.7.4).

6.3.5 Flow distribution. A certification of compliance confirming that the hot water generator should have sufficient internal water flow distribution to protect all tubes and prevent steam formation throughout the entire range of required operating condition should be provided to the contractor officer (see 3.9.3).

6.3.6 Remote control panel installation. When a free-standing remote mounted control panel is to be furnished (see 3.15), installation instructions covering the connection of the control panel should be provided.

6.3.7 Relief valve. A certificate of compliance verifying the performance and results of therelief valve pressure test (see 4.6) should be provided to the contracting officer.

6.3.8 Technical manuals. The manufacturer should provide two commercial technical manuals normally prepared and supplied with the boiler.

6.3.9 Boiler operational certification (Navy Only). The boiler operational certification should be performed, where applicable, per NAVFAC MO-324, after the boiler complies with the on-site tests (see 4.7). The boiler inspector should be certified by NAVFAC and licensed by the Engineering Field Division.

6.4 Definitions. Definitions used in this specification have been in accordance with the ABMA Lexicon of Boiler and Auxiliary Equipment. Other terms used in this document should be interpreted in accordance with the referenced national standards.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:38Z
Check the source to verify that this is the current version before use.

6.5 Subject term (key word) listing.

Coal fired
Gas and oil fired
High and low pressure
Hot water generator

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review Activities:
Army - MD
Air force - 84

User Activities:
Navy - MC
Army - ME

Preparing activity:
Navy - YD

(Project 4410-0073)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:38Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-B-28686	900911

3. DOCUMENT TITLE Boilers, Hot Water, Watertube, (Bent Tube, Multi-Drum and Cross Drum) Packaged Type (320,001 to 125,000,000 Btu/hr Thermal Output Capacity)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
D. C. MUI, Project Manager Facilities Criteria Development Divison	(805) 982-5801 ext 424	551-5801 ext 424
c. ADDRESS (Include Zip Code) Commanding Officer (156) Naval Construction Battalion Center Port Hueneme, CA 93043-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:38Z
Check the source to verify that this is the current version before use.